EXPERIMENT 4

SUPERHETERODYNE RECEIVER OF AM

```
clc; clear; close all;

% Parameters fs = 100000; t = 0:1/fs:0.02; fm = 500; % Message frequency fc = 10000; % Carrier frequency fLO = 14550; % Local Oscillator (IF = 4550 Hz)

% Message Signal m = cos(2pifm*t);

% Carrier Signal c = cos(2pifc*t);

% AM Signal am = (1 + m).cos(2pifct);

% Local Oscillator LO = cos(2pifLO*t);

% Frequency Mixing mixed = am .* LO;

% IF Filter [b,a] = butter(4,5000/(fs/2)); IF = filter(b,a,mixed);

% Envelope Detection env = abs(hilbert(IF));

% Display figure;

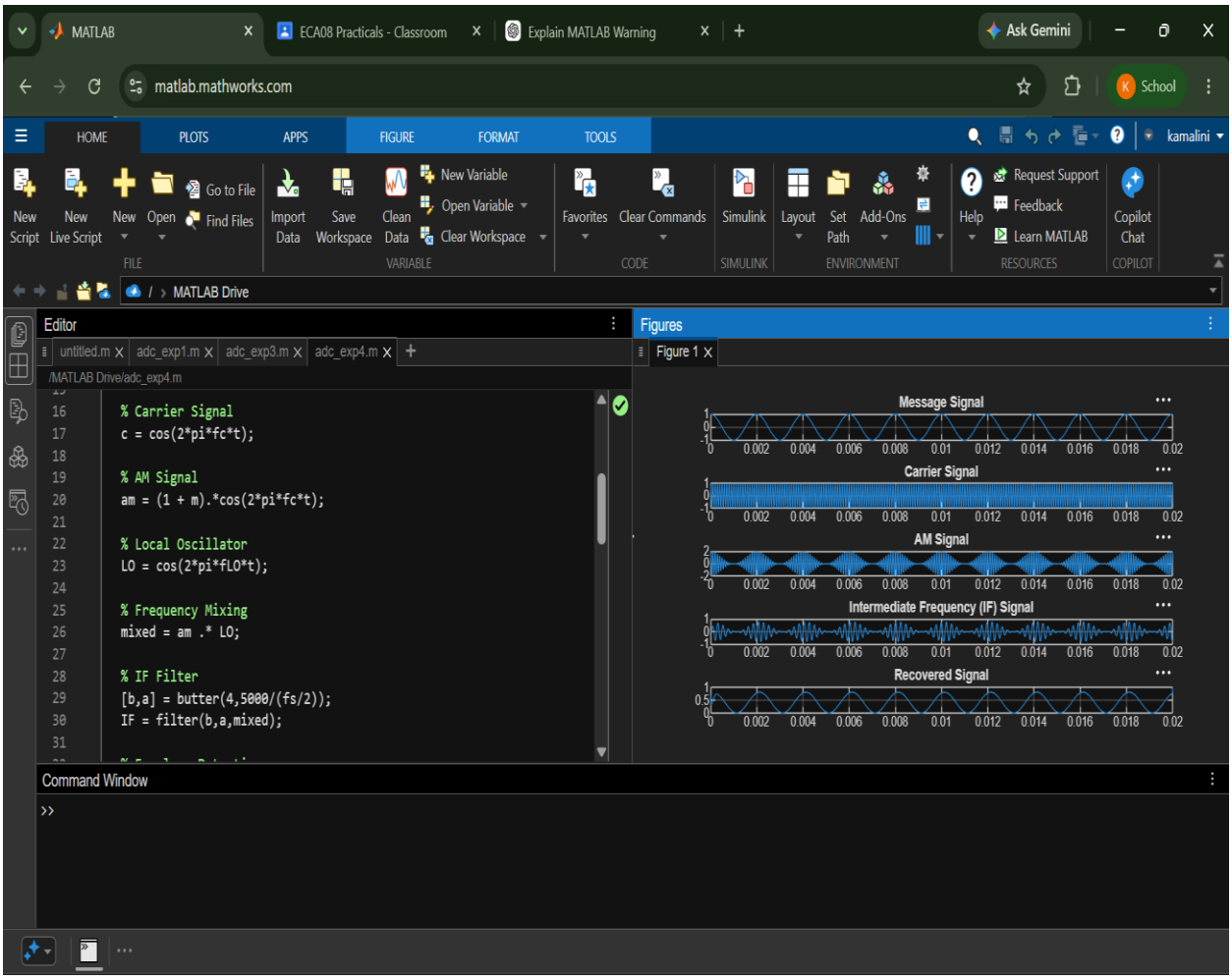
subplot(5,1,1) plot(t,m) title('Message Signal') grid on

subplot(5,1,2) plot(t,c) title('Carrier Signal') grid on

subplot(5,1,3) plot(t,am) title('AM Signal') grid on

subplot(5,1,4) plot(t,IF) title('Intermediate Frequency (IF) Signal') grid on

subplot(5,1,5) plot(t,env) title('Recovered Signal') grid on
```


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Copilot Chat
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Editor
untitled.m
adc_exp1.m
adc_exp3.m
adc_exp4.m
/MATLAB Drive/adc_exp4.m
% Carrier Signal
c = cos(2*pi*fc*t);
% AM Signal
am = (1 + m).*cos(2*pi*fc*t);
% Local Oscillator
LO = cos(2*pi*fLO*t);
% Frequency Mixing
mixed = am .* LO;
% IF Filter
[b,a] = butter(4,5000/(fs/2));
IF = filter(b,a,mixed);
Figures
Figure 1
Message Signal
Carrier Signal
AM Signal
Intermediate Frequency (IF) Signal
Recovered Signal
0 0.002 0.004 0.006 0.008 0.01 0.012 0.014 0.016 0.018 0.02
Command Window
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